

Chirag Venkatesh

Residence: Bangalore, India

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Title: Electronics & Communication Engineer

Technical Skills

Languages	Python, C, MATLAB
Frameworks & Tools	n8n, Qt, Plotly, NumPy, Pandas, MQTT
Systems & Dev Tools	Git, Wireshark, Linux, LTSpice, Cadence Virtuoso, Vivado
CS Fundamentals	Data Structures & Algorithms, Operating Systems, OOP, DBMS, Software Engineering

Professional Experience

OpenFabric (Decentralized AI Infrastructure) <i>Project Consultant</i>	<i>May 2025 – Jul 2025</i> <i>Bangalore, India</i>
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- Designed and developed a modular high-frequency cryptocurrency trading system using Python, deployed on a decentralized AI compute network.
- Implemented momentum and Bollinger Bands mean-reversion strategies, achieving a Sharpe ratio of 2.4 and high backtest-to-live consistency.
- Built multithreaded pipelines for live market data ingestion using NumPy and Pandas with optimized batch processing.
- Developed a real-time analytics dashboard using Plotly to visualize trade execution, P&L, and performance metrics.
- Improved strategy efficiency by 45% over baseline and reduced compute usage by 30%.

Bharat Earth Movers Ltd. (BEML) — Rail & Metro Division <i>Software Engineer</i>	<i>Aug 2024 – Nov 2024</i> <i>Bangalore, India</i>
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- Led development for Train Control Monitoring Systems with a focus on real-time software visualization.
- Built GUI applications using the Qt framework for monitoring operational and control parameters.
- Integrated Local Interconnect Network (LIN) communication for efficient, low-latency auxiliary systems.
- Collaborated with cross-functional teams to debug, test, and deploy software modules in production-grade systems.

Dassault Systèmes <i>Product Development Intern</i>	<i>May 2022 – Jul 2023</i> <i>Bangalore, India</i>
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- Designed and simulated automated systems using Modelica and electrical system design tools.
- Performed circuit simulation, validation, and performance optimization under real-world constraints.
- Worked closely with stakeholders on requirements gathering, system validation, and cost optimization.

Projects

Readimentary (RSVP PDF Reader) 

- Built a browser-based RSVP reader with **ORP-centered word rendering** and adjustable WPM for focus-optimized reading.
- Implemented chapter detection and persistent local library using **IndexedDB** and **localStorage**.
- Developed navigation controls with progress memory, jump-to-section, and chapter resume logic.
- Added UI state management for pause/resume, reading history, and interactive session controls.
- Delivered a full in-browser flow with **React 19**, **Vite 7**, **Tailwind CSS**, and **PDF.js**.

Sneaki — AI-Powered Activity to Google Calendar 🦊

- Built an automated system that logs application/window activity into Google Calendar with AI-based task categorization.
- Processes webhook activity payloads and extracts task name, category, and productivity score via JSON-returning AI backends.
- Creates structured calendar events with computed durations and metadata (app name, URL, task, score).
- Supports local or hosted AI endpoints and resilient handling of missing data.

Statistical Aerboreal Bird Repeller 🦊

- Built a computer vision pipeline using **SIFT** and **HOG** descriptors for robust bird detection under scale and lighting changes.
- Compared live video frames against a labeled dataset of bird species images to compute similarity scores.
- Designed a density estimator that aggregates detections over time to produce stable control signals.
- Mapped density scores to **PWM duty cycles** for dynamic ultrasonic actuation instead of fixed bursts.
- Documented results and field testing in a **peer-reviewed conference paper**.

Real-Time Gesture-Controlled Robotic Hand 🦊

- Built a mechanical system mirroring hand movements in **real time** with low-latency feedback.
- Implemented **Python-based signal processing** for sensor filtering, smoothing, and gesture stability.
- Used **MQTT** messaging to stream and synchronize servo state updates across devices.
- Tuned motor response curves to reduce jitter and improve fine-grained positional accuracy.

Education

PES University
B.Tech in Electronics and Communication Engineering
Minor in Computer Science Engineering

Bangalore, India
2020 – 2024